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**Supplementary information**

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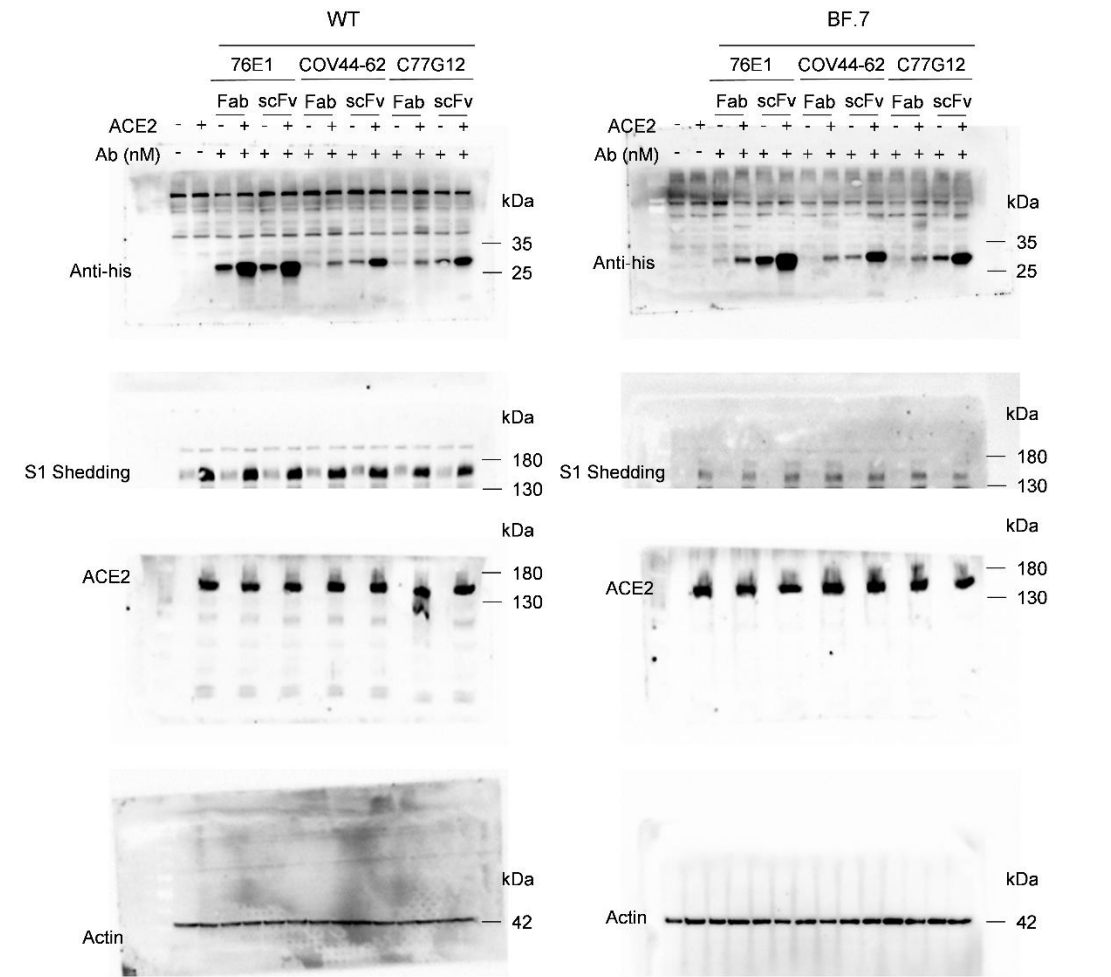
**Steric hindrance of antibody binding in an  
Omicron spike fusion intermediate**

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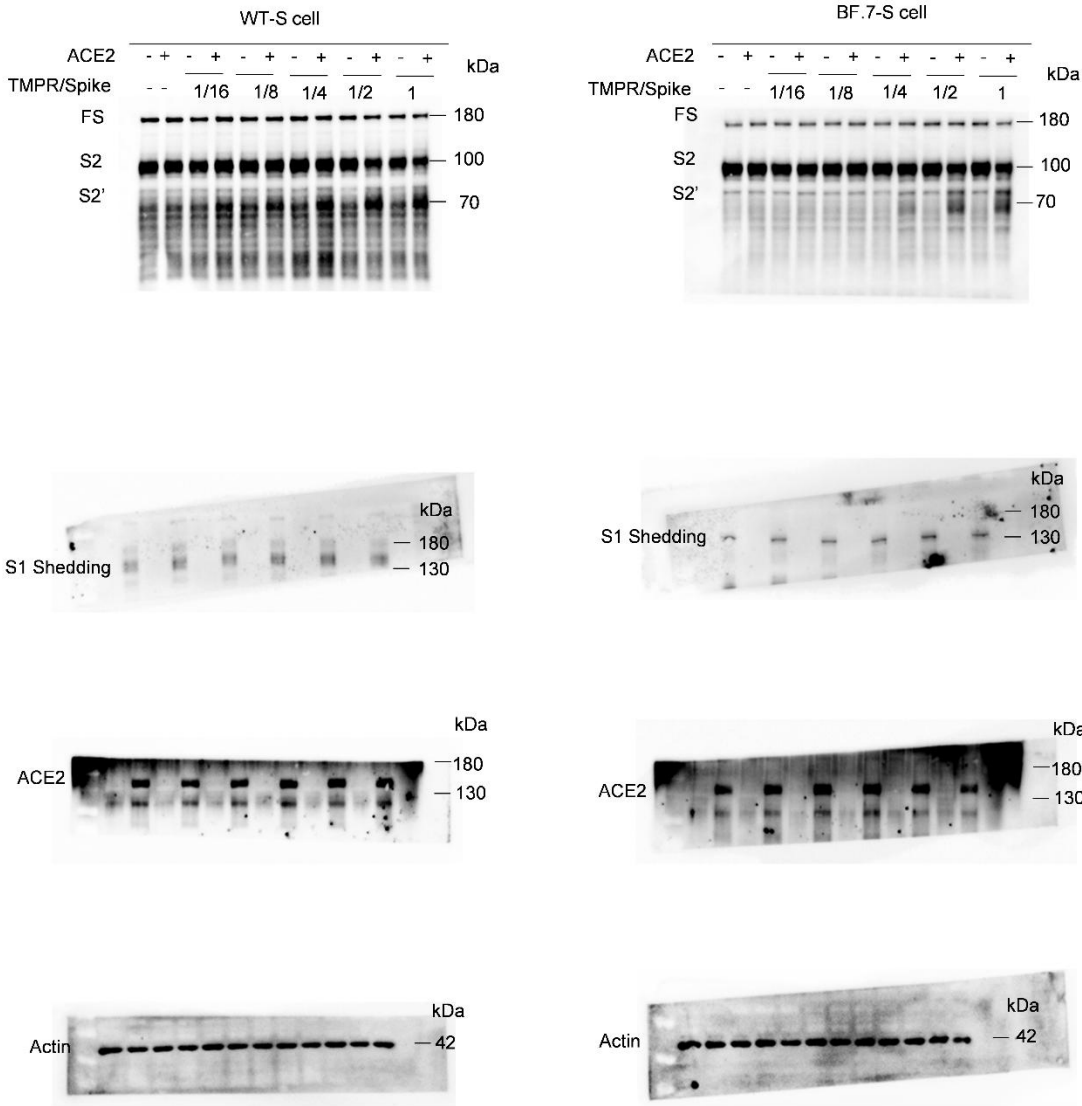
In the format provided by the  
authors and unedited

**Supplementary Figure 1** Uncropped gel images. Beta-actin were run on separate gels as sample processing controls

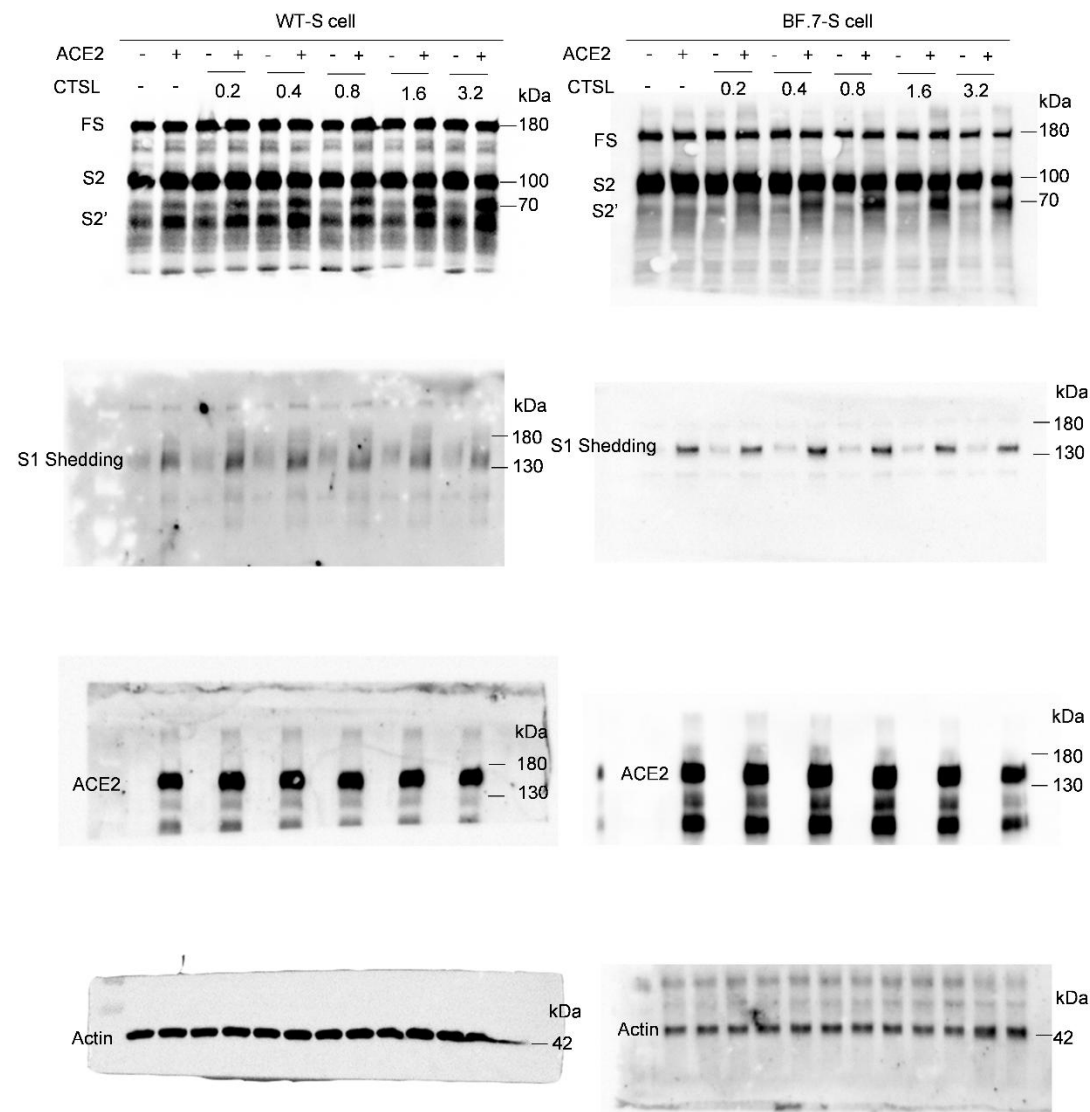
For Fig. 1e



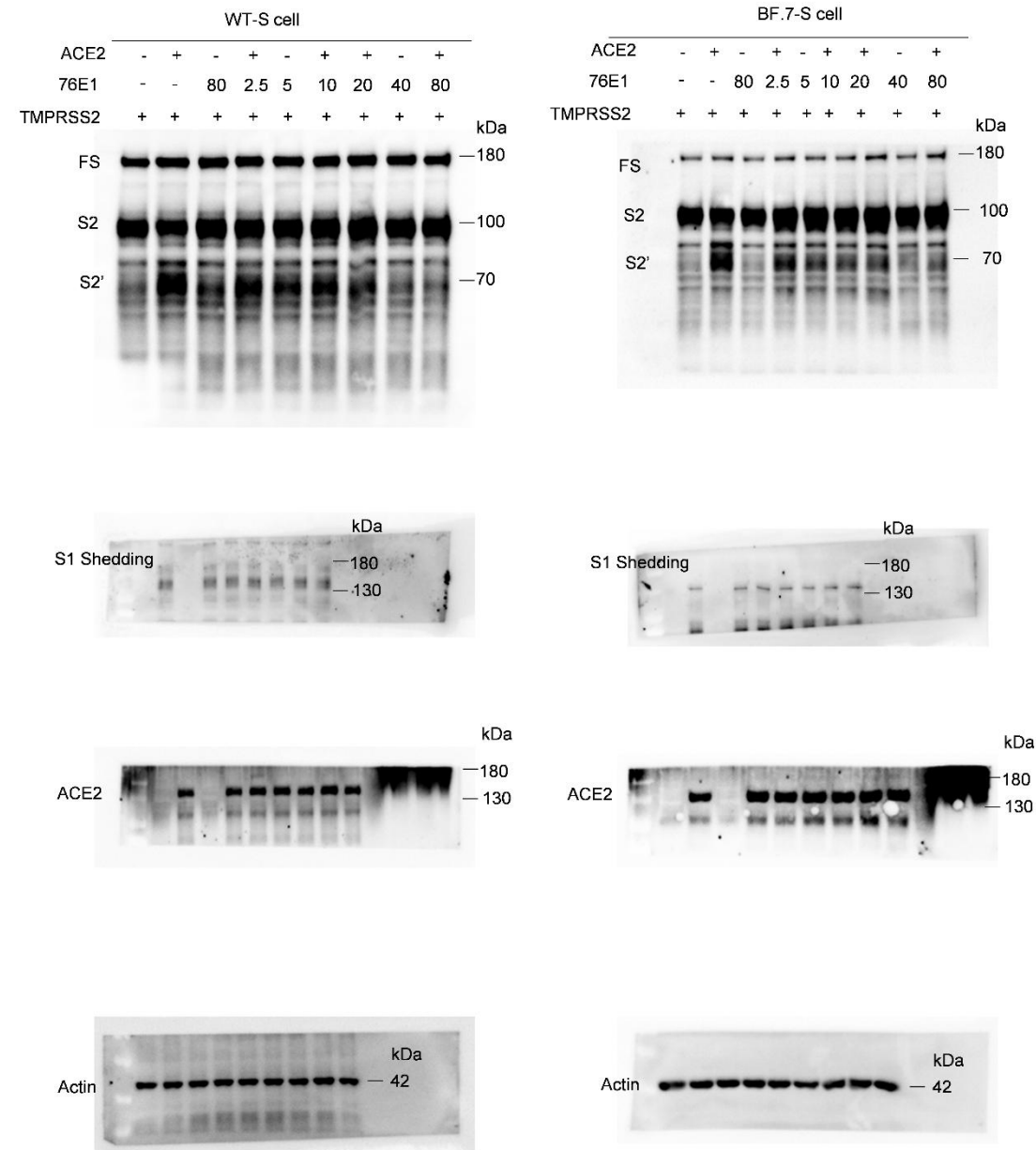
For Fig. 3a



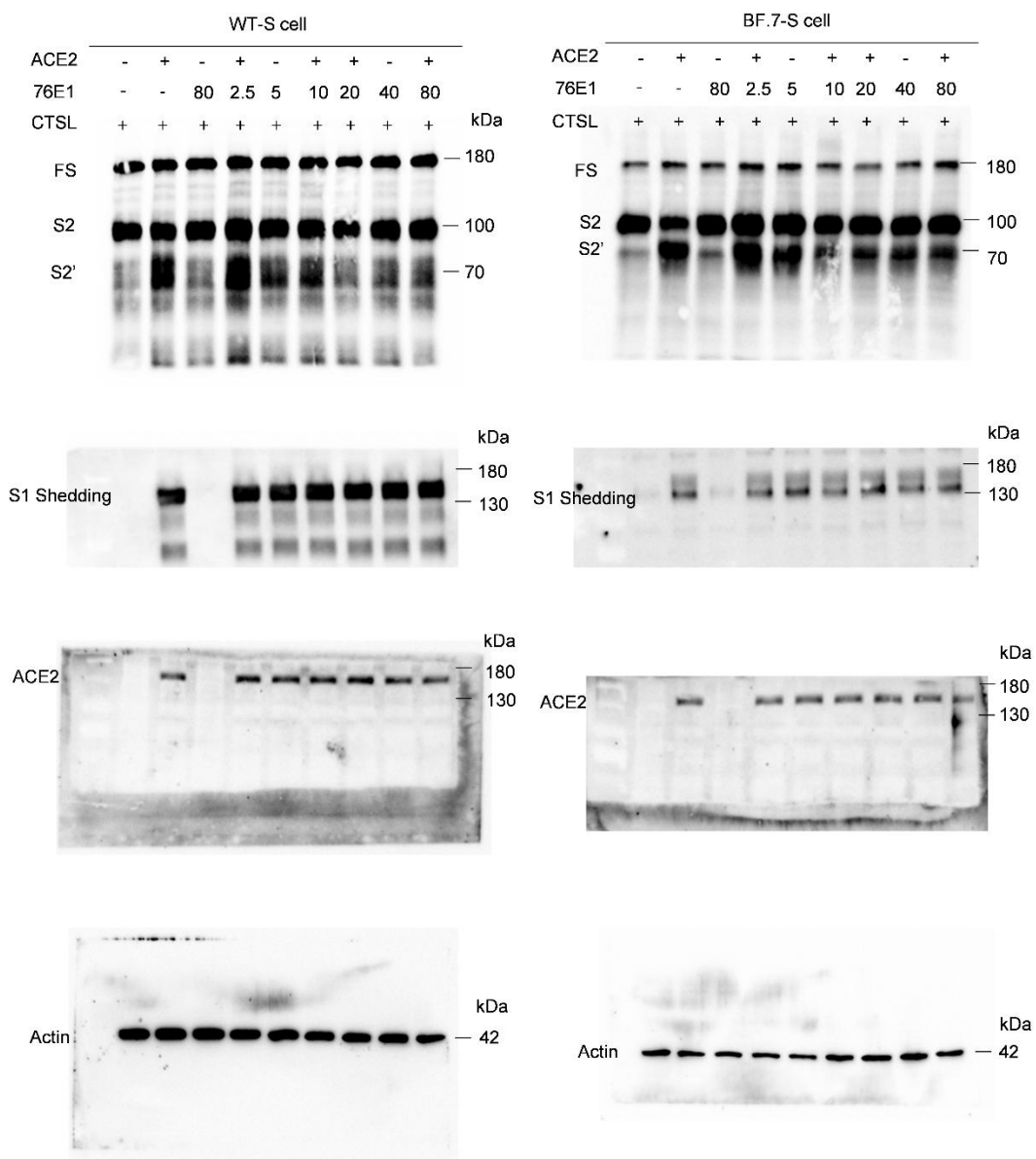
For Fig. 3b



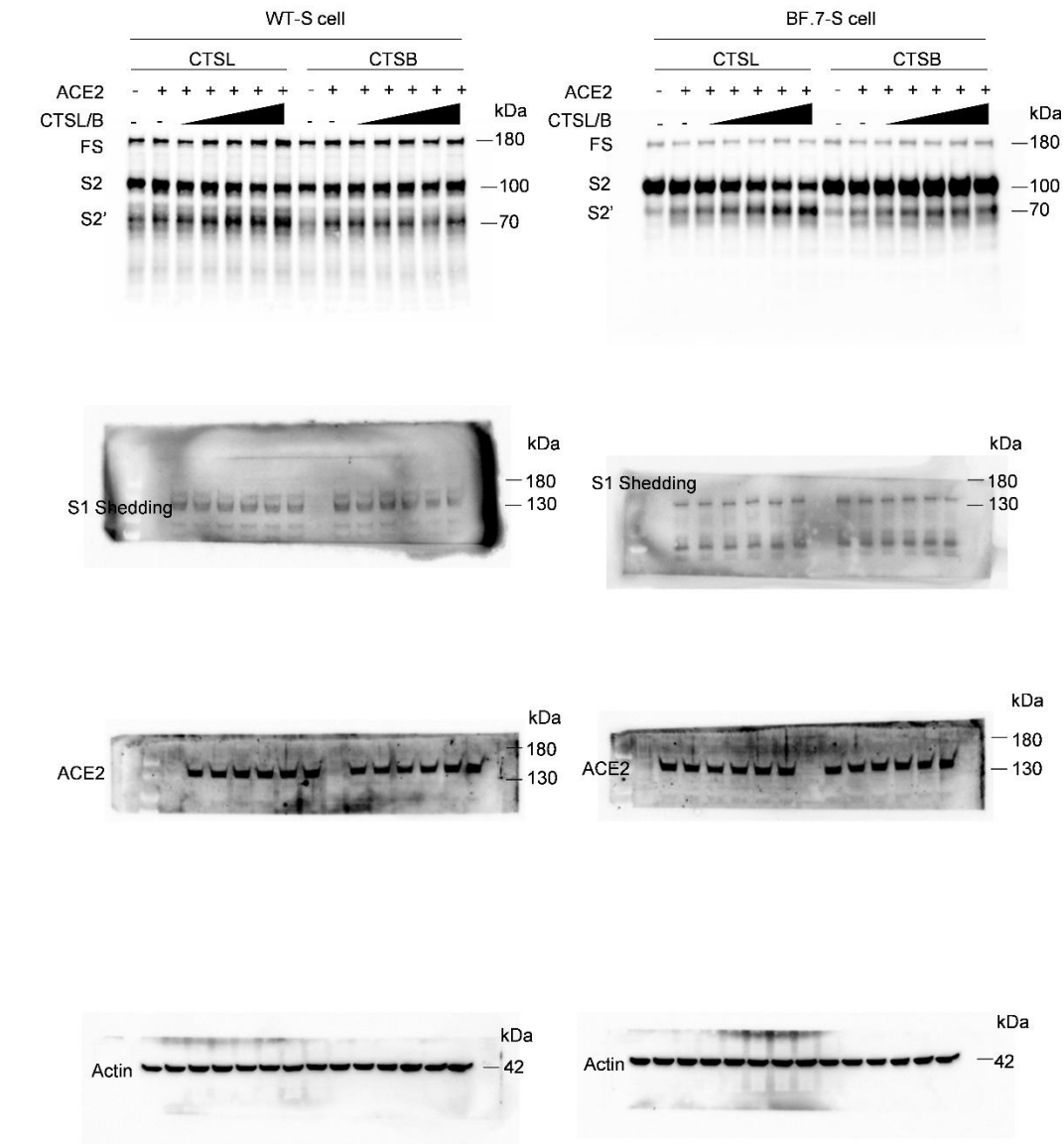
For Fig. 3c



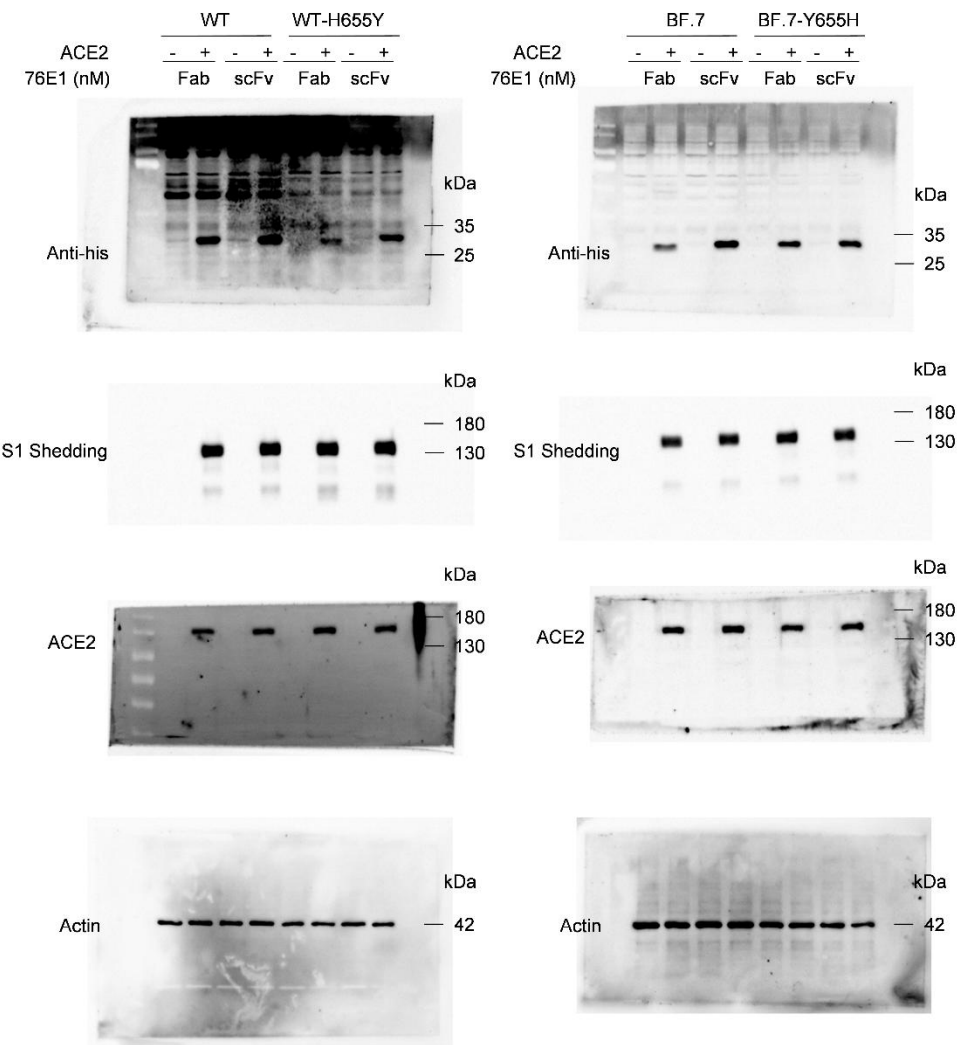
For Fig. 3d



For Extended Data Fig. 7

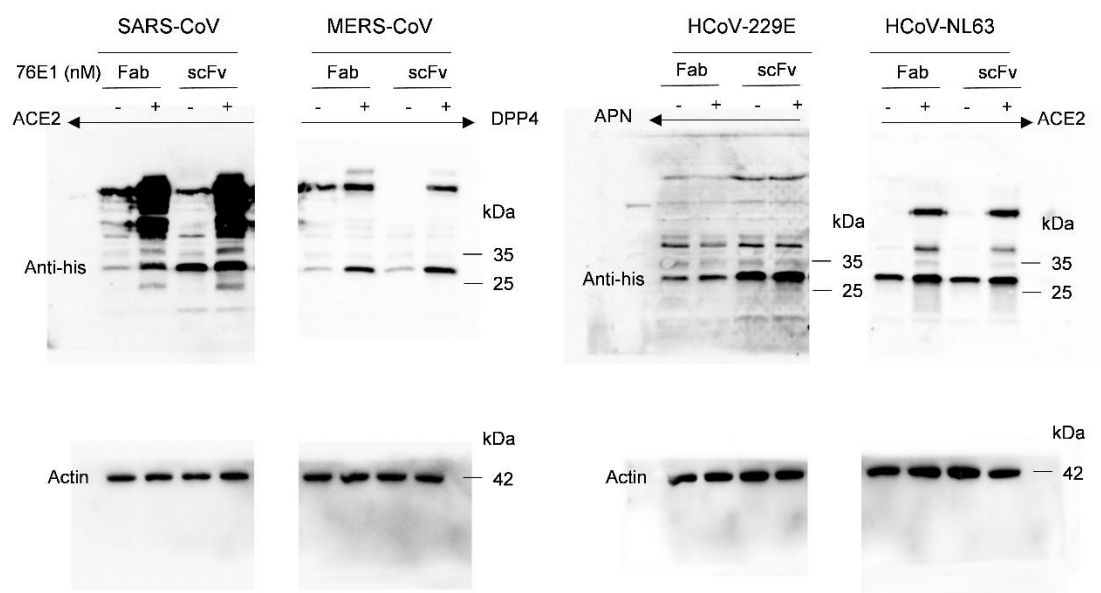


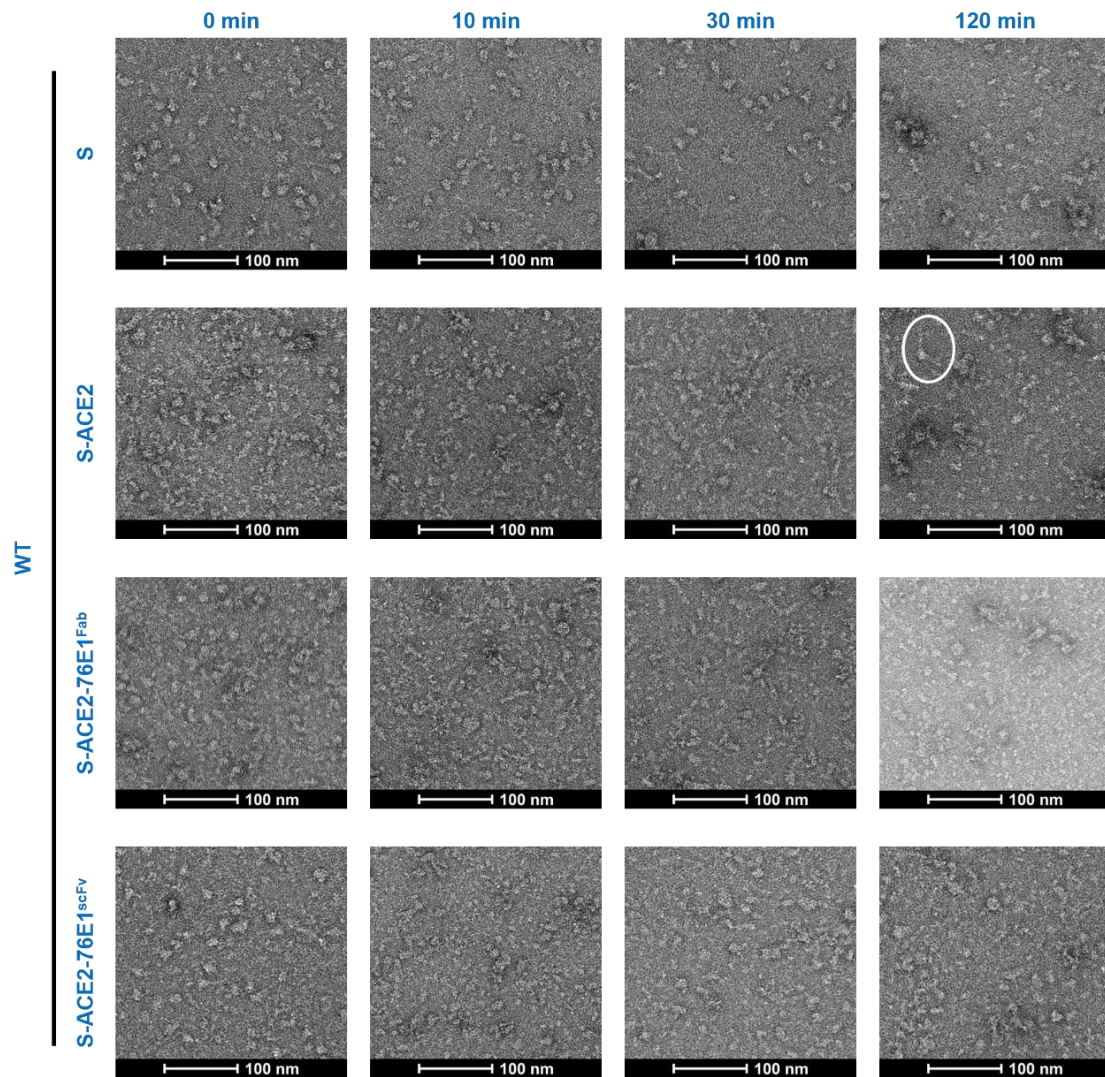
For Fig. 4c



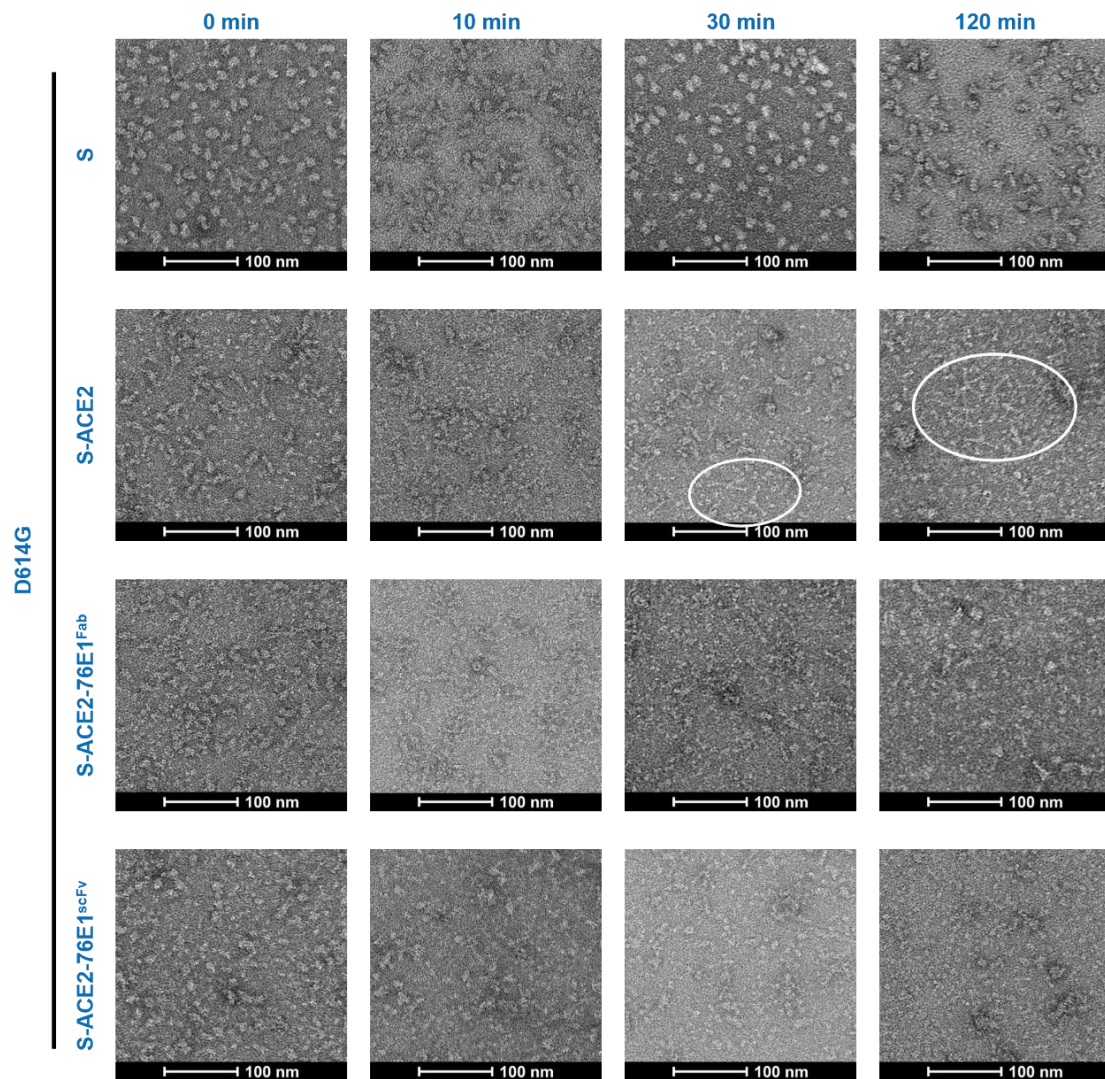


For Fig. 5d



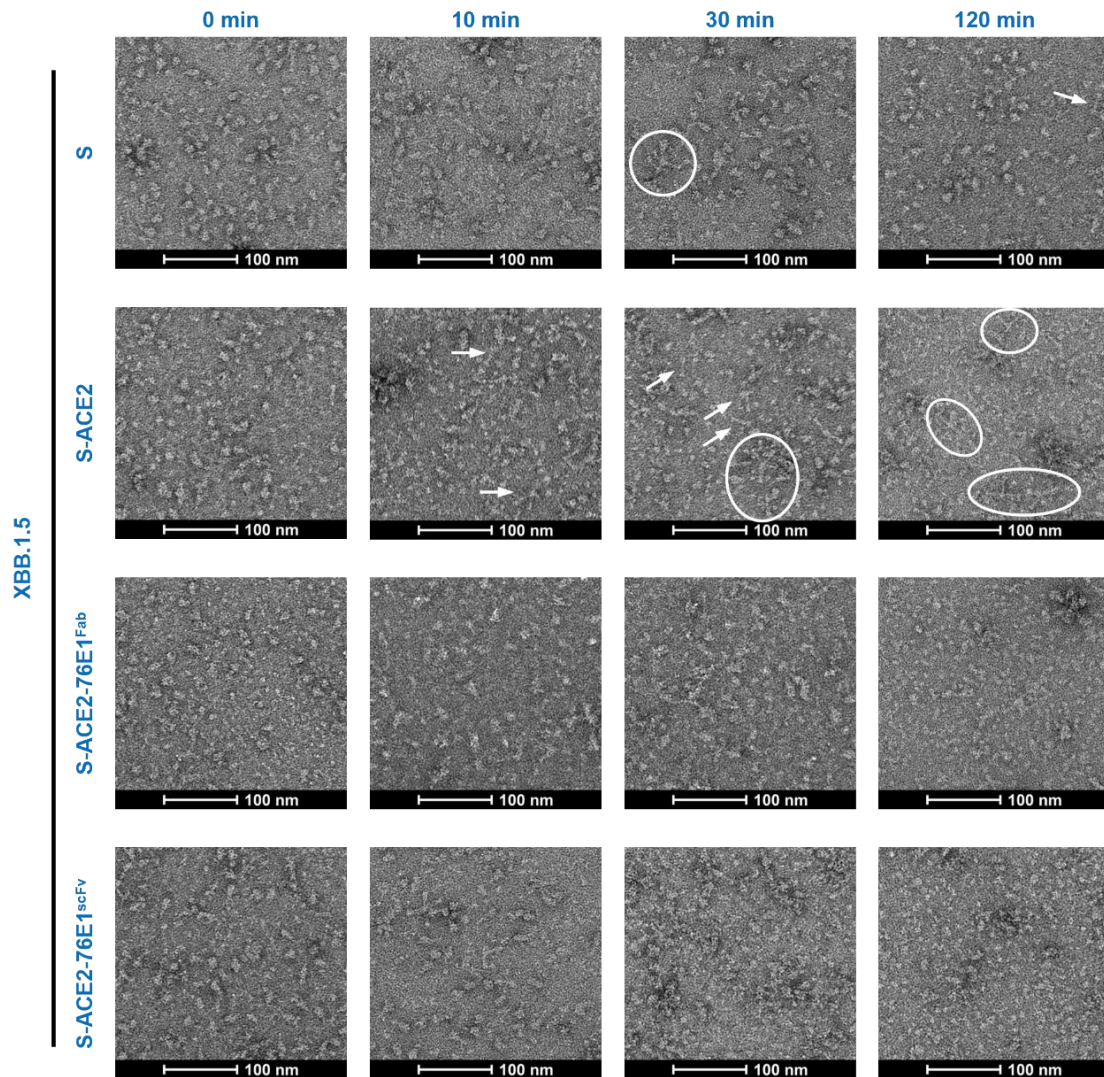


**Supplementary Figure 2. Impact of varying sizes of 76E1 on the WT S transition after ACE2 priming.** The 76E1-Fab and scFv could effectively block the transition of ACE2-primed WT S to the postfusion conformation.

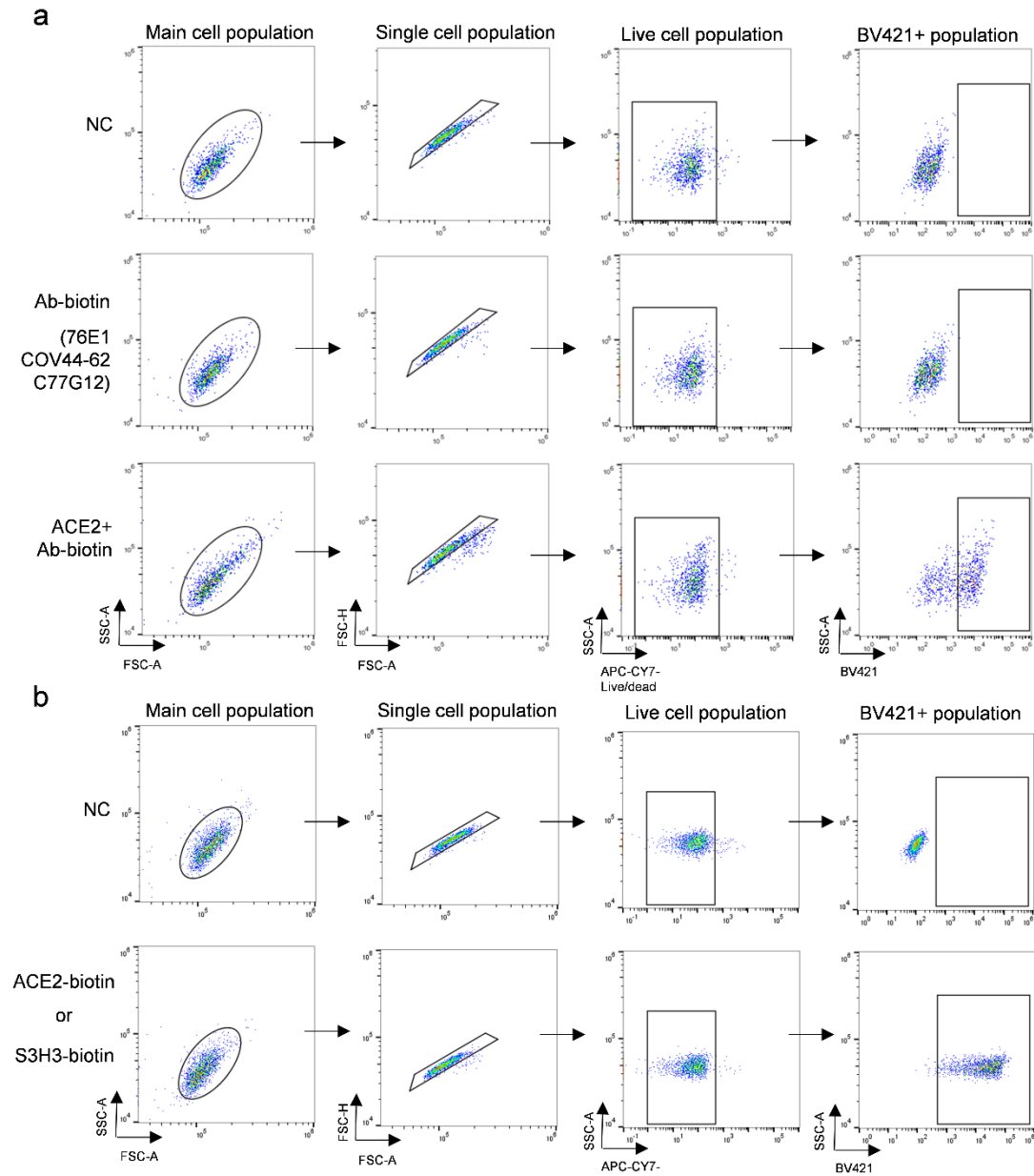


**Supplementary Figure 3. Impact of varying sizes of 76E1 on the D614G S transition after ACE2 priming.** ACE2-activated D614G S could rapidly form a postfusion conformation, which might be hindered by 76E1-Fab and scFv.



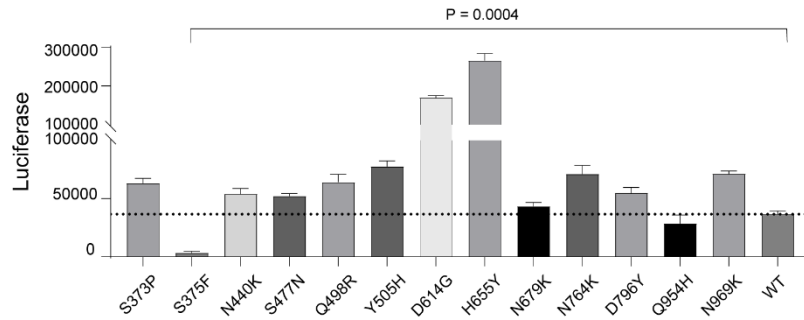


**Supplementary Figure 4. Impact of varying sizes of 76E1 on the XBB.1.5 S transition after ACE2 priming.** The XBB.1.5 S was less stable and ACE2-activated XBB.1.5 S could rapidly form a postfusion conformation. Few postfusion particles were observed in the mixtures containing 76E1-Fab and scFv, comparison to the S-ACE2 mixture.

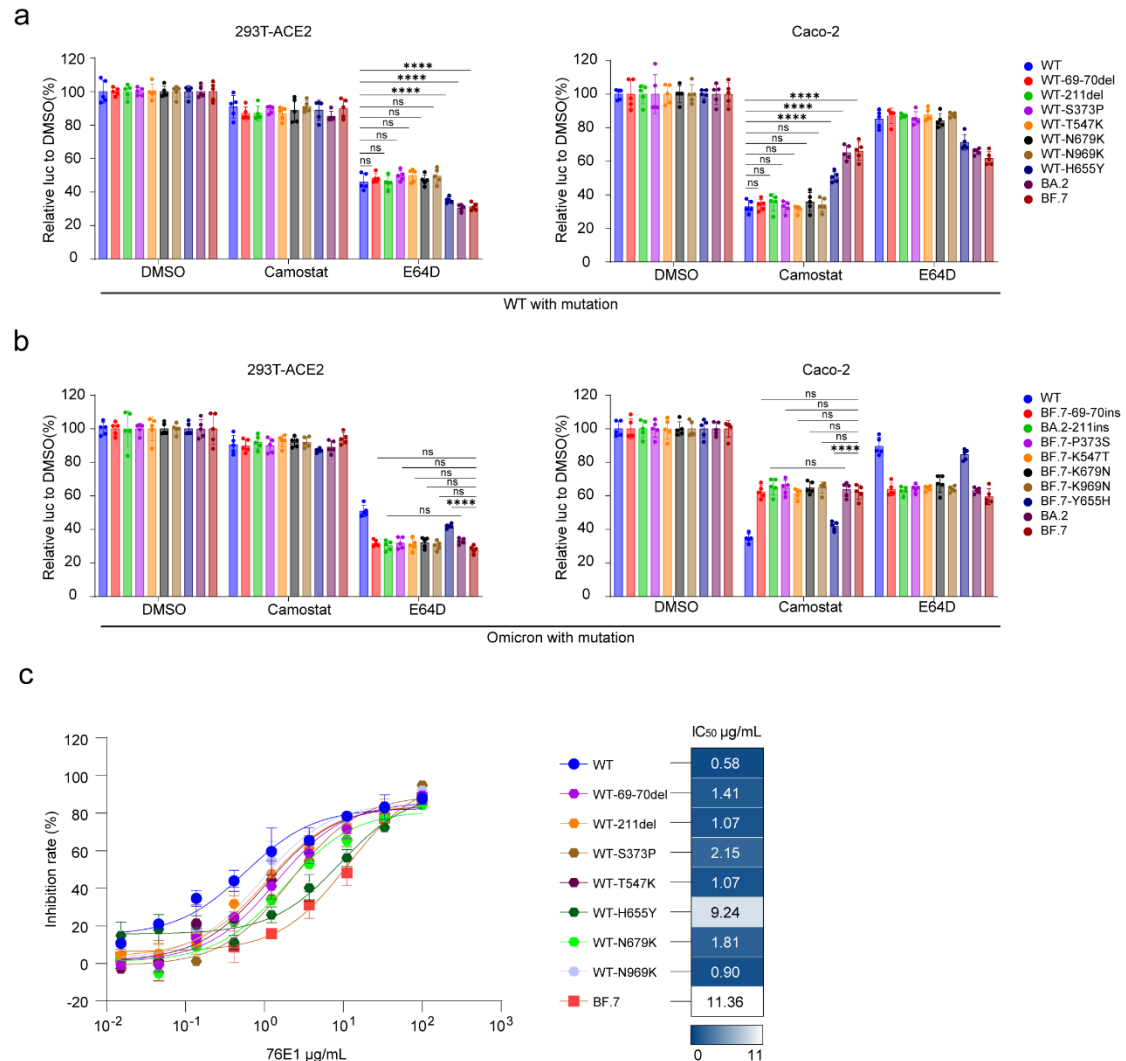


**Supplementary Figure 5. Flow cytometry gating strategies.**

**a**, Representative flow cytometry gating strategies related to Fig.1c (for Ab 76E1, COV44-62 and C77G12) and Extended Data Fig. 8c (for Ab 76E1). **b**, Representative flow cytometry gating strategies related to Fig.1c (for Ab S3H3) and Extended Data Fig. 8d (for ACE2). Live cell gating was omitted in Extended Data Fig. 8d due to low pH-induced cell death.



**Supplementary Figure 6. The infectivity of SARS-CoV-2 pseudoviruses harboring individual mutation as compared with the WT pseudovirus was conducted in 293T-ACE2 cells.** The S375F mutation resulted in a near-complete loss of pseudovirus infectivity.  $n = 3$ . The P values are analyzed by two-sided unpaired t tests. \*\*\* $P < 0.001$ . Data are shown as mean  $\pm$  SEM. Data representative of 2 experiments.



**Supplementary Figure 7. Comparison of H655Y with other reported mutations that modestly influence viral entry pathway.** Mutations 69-70ins, 211ins, P373S, T547K, K679N, and K969N have been previously reported to subtly affect viral entry route; here their impact on pathway choice and 76E1 neutralization is compared with that of H655Y. **a-b**, Relative infection of pseudoviruses carrying the indicated mutations in WT (**a**) or Omicron-reverted (**b**) backgrounds after treatment with 10  $\mu\text{M}$  Camostat or E64d in 293T-ACE2 or Caco-2 cells. Parental WT, BA.2, or BF.7 pseudoviruses were used as controls according to the strain background of each mutation (211ins present in BA.2; others in BF.7). **c**, Neutralization IC<sub>50</sub> of 76E1-FL against the mutant pseudoviruses in 293T-ACE2 cells. Data are mean  $\pm$  SEM ( $n=3$ ); two-sided unpaired t-tests were used to calculate  $P$  values. \*\*\*\*,  $P \leq 0.0001$ .

**Supplementary Table 1. Cryo-EM data collection, refinement and validation statistics of the WT S<sup>E-FIC</sup>-76E1 structure.**

|                                                     | S <sup>E-FIC</sup> -76E1<br>(EMDB-63002)<br>(PDB 9LDJ) | Local S2-76E1<br>(EMDB-63000)<br>(PDB 9LD2) | Local S2-76E1 top<br>EMDB-62992 |
|-----------------------------------------------------|--------------------------------------------------------|---------------------------------------------|---------------------------------|
| <b>Data collection and processing</b>               |                                                        |                                             |                                 |
| Magnification                                       | 130,000                                                | 130,000                                     | 130,000                         |
| Voltage (kV)                                        | 300                                                    | 300                                         | 300                             |
| Electron exposure (e <sup>-</sup> /Å <sup>2</sup> ) | 50                                                     | 50                                          | 50                              |
| Defocus range (μm)                                  | -1 to -3                                               | -1 to -3                                    | -1 to -3                        |
| Pixel size (Å)                                      | 0.932                                                  | 0.932                                       | 0.932                           |
| Symmetry imposed                                    | C3                                                     | C3                                          | C3                              |
| Initial particle images (no.)                       | 347,474                                                | 347,474                                     | 347,474                         |
| Final particle images (no.)                         | 93,138                                                 | 93,138                                      | 33,190                          |
| Map resolution (Å)                                  | 6.21                                                   | 3.73                                        | 3.88                            |
| FSC threshold                                       | 0.143                                                  | 0.143                                       | 0.143                           |
| <b>Refinement</b>                                   |                                                        |                                             |                                 |
| Initial model used (PDB code)                       |                                                        | 8Z7P, 7X9E                                  |                                 |
| Model resolution (Å)                                |                                                        | 3.73                                        |                                 |
| FSC threshold                                       |                                                        | 0.143                                       |                                 |
| Map sharpening B factor (Å <sup>2</sup> )           | -474.6                                                 | -108.7                                      | -131.7                          |
| Model composition                                   |                                                        |                                             |                                 |
| Non-hydrogen atoms                                  |                                                        | 19,800                                      |                                 |
| Protein residues                                    |                                                        | 2,583                                       |                                 |
| Ligands                                             |                                                        | 18                                          |                                 |
| B factors (Å <sup>2</sup> )                         |                                                        |                                             |                                 |
| Protein                                             |                                                        | 138                                         |                                 |
| Ligand                                              |                                                        | 180                                         |                                 |
| R.m.s. deviations                                   |                                                        |                                             |                                 |
| Bond lengths (Å)                                    |                                                        | 0.002                                       |                                 |
| Bond angles (°)                                     |                                                        | 0.410                                       |                                 |
| Validation                                          |                                                        |                                             |                                 |
| MolProbity score                                    |                                                        | 2.16                                        |                                 |
| Clashscore                                          |                                                        | 6.80                                        |                                 |
| Poor rotamers (%)                                   |                                                        | 4.12                                        |                                 |
| Ramachandran plot                                   |                                                        |                                             |                                 |
| Favored (%)                                         |                                                        | 95.47                                       |                                 |
| Allowed (%)                                         |                                                        | 4.53                                        |                                 |
| Disallowed (%)                                      |                                                        | 0.00                                        |                                 |



**Supplementary Table 2. Cryo-EM data collection, refinement and validation statistics of the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure.**

|                                                     | S <sup>E-FIC</sup> -76E1<br>(EMDB-65168)<br>(PDB 9VLT) | Local S2-76E1<br>(EMDB-65166)<br>(PDB 9VLS) | Local S2-76E1 top<br>EMDB-65164 |
|-----------------------------------------------------|--------------------------------------------------------|---------------------------------------------|---------------------------------|
| <b>Data collection and processing</b>               |                                                        |                                             |                                 |
| Magnification                                       | 130,000                                                | 130,000                                     | 130,000                         |
| Voltage (kV)                                        | 300                                                    | 300                                         | 300                             |
| Electron exposure (e <sup>-</sup> /Å <sup>2</sup> ) | 50                                                     | 50                                          | 50                              |
| Defocus range (μm)                                  | -1 to -3                                               | -1 to -3                                    | -1 to -3                        |
| Pixel size (Å)                                      | 0.932                                                  | 0.932                                       | 0.932                           |
| Symmetry imposed                                    | C3                                                     | C3                                          | C3                              |
| Initial particle images (no.)                       | 154,469                                                | 154,469                                     | 154,469                         |
| Final particle images (no.)                         | 93,565                                                 | 39,788                                      | 29,151                          |
| Map resolution (Å)                                  | 5.65                                                   | 4.14                                        | 4.14                            |
| FSC threshold                                       | 0.143                                                  | 0.143                                       | 0.143                           |
| <b>Refinement</b>                                   |                                                        |                                             |                                 |
| Initial model used (PDB code)                       |                                                        | 8VKO, 7X9E                                  |                                 |
| Model resolution (Å)                                |                                                        | 4.14                                        |                                 |
| FSC threshold                                       |                                                        | 0.143                                       |                                 |
| Map sharpening B factor (Å <sup>2</sup> )           | -271.0                                                 | -112.1                                      | -131.7                          |
| Model composition                                   |                                                        |                                             |                                 |
| Non-hydrogen atoms                                  |                                                        | 19,572                                      |                                 |
| Protein residues                                    |                                                        | 2,559                                       |                                 |
| Ligands                                             |                                                        | 12                                          |                                 |
| B factors (Å <sup>2</sup> )                         |                                                        |                                             |                                 |
| Protein                                             |                                                        | 138                                         |                                 |
| Ligand                                              |                                                        | 224                                         |                                 |
| R.m.s. deviations                                   |                                                        |                                             |                                 |
| Bond lengths (Å)                                    |                                                        | 0.002                                       |                                 |
| Bond angles (°)                                     |                                                        | 0.434                                       |                                 |
| Validation                                          |                                                        |                                             |                                 |
| MolProbity score                                    |                                                        | 2.31                                        |                                 |
| Clashscore                                          |                                                        | 6.66                                        |                                 |
| Poor rotamers (%)                                   |                                                        | 5.14                                        |                                 |
| Ramachandran plot                                   |                                                        |                                             |                                 |
| Favored (%)                                         |                                                        | 93.99                                       |                                 |
| Allowed (%)                                         |                                                        | 6.01                                        |                                 |
| Disallowed (%)                                      |                                                        | 0.00                                        |                                 |

**Supplementary Table 3. Summary of pseudovirus neutralizing IC<sub>50</sub> values from three biological replicates. Related to Fig. 1b.**

| IC <sub>50</sub> (µg/mL) |                   | WT                | Delta            | BA.2              | BA.5              | BF.7              | BQ.1.1            | XBB.1.5           |
|--------------------------|-------------------|-------------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| <b>76E1</b>              | Exp 1             | 0.32              | 1.45             | 7.89              | 9.44              | 19.47             | 11.38             | 15.54             |
|                          | Exp 2             | 0.65              | 1.06             | 5.34              | 12.62             | 20.32             | 17.91             | 16.04             |
|                          | Exp 3             | 0.77              | 1.02             | 7.67              | 9.07              | 16.16             | 11.49             | 15.70             |
|                          | <b>Mean ± SEM</b> | <b>0.58±0.14</b>  | <b>1.17±0.14</b> | <b>6.96±0.82</b>  | <b>10.38±1.13</b> | <b>18.65±1.27</b> | <b>13.59±2.16</b> | <b>15.76±0.15</b> |
| <b>COV44-62</b>          | Exp 1             | 4.48              | 5.95             | 47.43             | 23.70             | 52.65             | 41.52             | 56.17             |
|                          | Exp 2             | 6.02              | 6.22             | 37.12             | 20.68             | 43.79             | 53.34             | 59.08             |
|                          | Exp 3             | 7.15              | 8.64             | 40.48             | 17.27             | 41.48             | 37.03             | 45.90             |
|                          | <b>Mean ± SEM</b> | <b>5.88±0.77</b>  | <b>6.94±0.85</b> | <b>41.68±3.04</b> | <b>20.55±1.86</b> | <b>45.97±3.40</b> | <b>43.96±4.86</b> | <b>53.72±4.00</b> |
| <b>C77G12</b>            | Exp 1             | 11.25             | 7.48             | 31.53             | 31.33             | 52.61             | 39.09             | 62.69             |
|                          | Exp 2             | 10.73             | 5.27             | 19.25             | 26.77             | 43.42             | 48.99             | 62.44             |
|                          | Exp 3             | 8.30              | 7.16             | 25.40             | 27.91             | 47.50             | 54.02             | 57.57             |
|                          | <b>Mean ± SEM</b> | <b>10.09±0.91</b> | <b>6.64±0.69</b> | <b>25.39±3.54</b> | <b>28.67±1.37</b> | <b>47.84±2.66</b> | <b>47.37±4.39</b> | <b>60.90±1.67</b> |

**Supplementary Table 4. Summary of pseudovirus neutralizing IC<sub>50</sub> values from two biological replicate. Related to Fig. 1g.**

| IC <sub>50</sub> (nM) |       |         | WT    | Delta | BA.2   | BA.5   | BF.7   | BQ.1.1 | XBB.1.5 |
|-----------------------|-------|---------|-------|-------|--------|--------|--------|--------|---------|
| 76E1                  | Exp 1 | FL      | 5.59  | 7.09  | 65.77  | 72.85  | 62.06  | 161.10 | 115.10  |
|                       |       | Fab     | 6.20  | 7.62  | 47.75  | 20.89  | 28.04  | 104.30 | 63.59   |
|                       |       | scFv    | 5.23  | 7.02  | 21.34  | 10.12  | 7.67   | 28.33  | 18.55   |
|                       | Exp 2 | FL      | 2.50  | 8.41  | 66.64  | 60.68  | 55.37  | 162.50 | 97.43   |
|                       |       | Fab     | 5.08  | 8.48  | 34.44  | 21.69  | 31.99  | 72.40  | 50.47   |
|                       |       | scFv    | 2.27  | 5.87  | 13.73  | 10.13  | 10.77  | 21.10  | 16.76   |
|                       | Mean  | FL      | 4.05  | 7.75  | 66.21  | 66.77  | 58.72  | 161.80 | 106.27  |
|                       |       | Fab     | 5.64  | 8.05  | 41.10  | 21.29  | 30.02  | 88.35  | 57.03   |
|                       |       | scFv    | 3.75  | 6.44  | 17.54  | 10.13  | 9.22   | 24.72  | 17.66   |
|                       |       | FL/Fab  | 0.72  | 0.96  | 1.61   | 3.14   | 1.96   | 1.83   | 1.86    |
|                       |       | FL/scFv | 1.08  | 1.20  | 3.78   | 6.59   | 6.37   | 6.55   | 6.02    |
| COV44-62              | Exp 1 | FL      | 34.74 | 52.10 | 148.50 | 874.90 | 262.40 | 316.30 | 559.50  |
|                       |       | Fab     | 34.11 | 52.70 | 50.94  | 375.60 | 103.50 | 49.64  | 227.50  |
|                       |       | scFv    | 7.64  | 14.23 | 13.52  | 40.26  | 28.12  | 15.42  | 33.82   |
|                       | Exp 2 | FL      | 33.42 | 41.46 | 87.50  | 756.30 | 226.10 | 321.90 | 633.50  |
|                       |       | Fab     | 36.59 | 41.53 | 51.07  | 237.00 | 90.46  | 56.48  | 218.60  |
|                       |       | scFv    | 9.52  | 11.14 | 8.42   | 43.21  | 34.06  | 19.34  | 43.36   |
|                       | Mean  | FL      | 34.08 | 46.78 | 118.00 | 815.60 | 244.25 | 319.10 | 596.50  |
|                       |       | Fab     | 35.35 | 47.12 | 51.01  | 306.30 | 96.98  | 53.06  | 223.05  |
|                       |       | scFv    | 8.58  | 12.69 | 10.97  | 41.74  | 31.09  | 17.38  | 38.59   |
|                       |       | FL/Fab  | 0.96  | 0.99  | 2.31   | 2.66   | 2.52   | 6.01   | 2.67    |
|                       |       | FL/scFv | 3.97  | 3.69  | 10.76  | 19.54  | 7.86   | 18.36  | 15.46   |
| C77G12                | Exp 1 | FL      | 48.37 | 35.15 | 309.10 | 408.30 | 269.20 | 472.60 | 465.50  |
|                       |       | Fab     | 52.51 | 31.11 | 162.80 | 330.00 | 65.06  | 248.60 | 120.80  |
|                       |       | scFv    | 19.65 | 13.98 | 74.52  | 159.50 | 28.74  | 40.89  | 62.00   |
|                       | Exp 2 | FL      | 47.88 | 31.14 | 294.30 | 402.00 | 289.20 | 421.90 | 465.70  |
|                       |       | Fab     | 41.26 | 32.92 | 129.60 | 301.80 | 107.60 | 121.00 | 247.10  |
|                       |       | scFv    | 19.76 | 15.13 | 63.95  | 133.10 | 47.33  | 43.38  | 51.08   |
|                       | Mean  | FL      | 48.37 | 33.15 | 301.70 | 405.15 | 279.20 | 447.25 | 465.60  |
|                       |       | Fab     | 46.89 | 32.02 | 146.20 | 315.90 | 86.33  | 184.80 | 183.95  |
|                       |       | scFv    | 19.71 | 14.56 | 69.24  | 146.30 | 38.04  | 42.14  | 56.54   |
|                       |       | FL/Fab  | 1.03  | 1.04  | 2.06   | 1.28   | 3.23   | 2.42   | 2.53    |
|                       |       | FL/scFv | 2.45  | 2.28  | 4.36   | 2.77   | 7.34   | 10.61  | 8.23    |

**Supplementary Table 5. Summary of pseudovirus neutralizing IC<sub>50</sub> values from two biological replicate. Related to Fig. 4d.**

| IC <sub>50</sub> (nM) |       |         | WT   | WT-H655Y | BF.7  | BF.7-Y655H |
|-----------------------|-------|---------|------|----------|-------|------------|
| 76E1                  | Exp 1 | FL      | 9.25 | 45.18    | 78.08 | 20.90      |
|                       |       | Fab     | 7.19 | 15.37    | 27.78 | 17.03      |
|                       |       | scFv    | 5.68 | 5.09     | 4.11  | 9.97       |
|                       | Exp 2 | FL      | 6.04 | 36.09    | 72.77 | 19.03      |
|                       |       | Fab     | 5.48 | 17.43    | 30.41 | 17.19      |
|                       |       | scFv    | 4.09 | 2.98     | 7.75  | 7.12       |
|                       | Mean  | FL      | 7.65 | 40.64    | 75.43 | 19.97      |
|                       |       | Fab     | 6.33 | 16.40    | 29.10 | 17.11      |
|                       |       | scFv    | 4.89 | 4.03     | 5.93  | 8.54       |
|                       |       | FL/Fab  | 1.21 | 2.48     | 2.59  | 1.17       |
|                       |       | FL/scFv | 1.57 | 10.07    | 12.72 | 2.34       |

**Supplementary Table 6. Summary of pseudovirus neutralizing IC<sub>50</sub> values from two biological replicate. Related to Fig. 4f.**

| IC <sub>50</sub> (nM) |       |      | WT        | WT-H655Y | BF.7  | BF.7-Y655H |       |
|-----------------------|-------|------|-----------|----------|-------|------------|-------|
| 76E1                  | Exp 1 | FL   | 293T-ACE2 | 5.22     | 52.97 | 93.03      | 22.62 |
|                       |       |      | Caco-2    | 6.70     | 13.21 | 12.44      | 10.20 |
|                       |       | scFv | 293T-ACE2 | 4.77     | 8.92  | 5.50       | 9.88  |
|                       |       |      | Caco-2    | 5.04     | 7.61  | 4.57       | 5.15  |
|                       | Exp 2 | FL   | 293T-ACE2 | 7.11     | 54.55 | 70.80      | 29.03 |
|                       |       |      | Caco-2    | 4.91     | 17.26 | 16.05      | 8.65  |
|                       |       | scFv | 293T-ACE2 | 6.74     | 8.29  | 5.67       | 8.69  |
|                       |       |      | Caco-2    | 5.95     | 8.64  | 2.63       | 5.72  |
|                       | Mean  | FL   | 293T-ACE2 | 6.17     | 53.76 | 81.92      | 25.83 |
|                       |       |      | Caco-2    | 5.81     | 15.24 | 14.25      | 9.43  |
|                       |       | scFv | 293T-ACE2 | 5.75     | 8.60  | 5.59       | 9.28  |
|                       |       |      | Caco-2    | 5.50     | 8.13  | 3.60       | 5.43  |